

POYRAZ KOCA

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ABOUT

Software Engineering graduate from TED University with an Honor Degree (GPA: 3.04/4.00). Experienced in full-stack, backend, and AI-driven software development, with a strong interest in Artificial Intelligence, Computer Vision, and Data Visualization.

EDUCATION

TED University – Bachelor of Science in Software Engineering
2022-2026 Ankara, Türkiye
Honor Degree (GPA: 3.04/4.00)

Göl Anadolu – High School
2019-2021 Kastamonu, Türkiye
GPA: 92.76/100

Seven Lakes – High School
2018-2019 Houston/TX, USA
Freshman Year

PROFESSIONAL EXPERIENCE

LotusIQ

Back-End Developer (Contract)
May 2026 – Present

IoT solutions provider specializing in KNX-based smart home automation systems.

- Developed and configured backend automation logic for KNX-based smart home systems.
- Implemented and maintained device communication and automation workflows for residential and commercial smart building solutions.
- Collaborated with engineers to integrate smart devices into scalable automation infrastructures.

TEDUApp

Back-End Developer (Part-Time)
September 2024 – February 2026

Official mobile application developed for the TED University community, providing academic, social, and campus services.

- Contributed to the backend development of the official TED University mobile application.
- Developed and maintained APIs supporting features such as course schedule management, campus event announcements, student clubs, department communities, and university podcasts.

- Collaborated with cross-functional teams to deliver scalable and reliable backend services for iOS and Android platforms.

Tech Stack: Flutter, FireBase, Cloud Firestore

([App Store](#) | [Google Play](#))

Coridor

Back-End Developer Intern

September 2025 – December 2025

Gamified social impact platform encouraging young people to participate in volunteering activities and convert earned points into donations and rewards.

- Developed backend services for a gamified social responsibility platform focused on youth engagement.
- Implemented server-side functionalities supporting user participation, reward management, and donation workflows.
- Collaborated with the development team to improve platform performance and maintain scalable backend architecture.

Tech Stack: Web API, Web Services

([Website](#) | [App Store](#) | [Google Play](#))

Outlier

Freelance Prompt Engineer

December 2024 – September 2025

AI training platform focused on improving large language model performance.

- Designed and optimized prompts to generate high-quality code across multiple programming languages, including Java, Python, C++, and Verilog.
- Evaluated AI-generated code for correctness, functionality, and adherence to software engineering best practices.
- Identified implementation errors and refined prompt strategies to improve model accuracy and code quality.

Tech Stack: Java, Go, Python

([Website](#))

Robot El Association

Data Reporting Analyst Intern

August 2024 – July 2025

Non-profit organization developing affordable 3D-printed prosthetic hands.

- Collected, analyzed, and reported project data to support organizational decision-making.
- Prepared analytical reports tracking project progress and identifying improvement opportunities based on participant feedback.

- Designed recommendations for standardized project data management practices.
- Conducted research on international statistical reporting methodologies to improve data collection processes beyond practices currently adopted in Türkiye.

Tech Stack: SSRS, R, SPSS, Microsoft Excel, Python

([Website](#))

Saat Teknoloji

Back-End Developer Intern

July 2024 – August 2024

Media technology company developing live TV and streaming platforms with EPG (Electronic Program Guide) technologies.

- Contributed to the backend development of a live streaming platform for TV channels, films, and series using Java Spring Boot.
- Integrated Elasticsearch to implement high-performance keyword-based search functionality.
- Assisted in developing and maintaining Electronic Program Guide (EPG) correction services to improve program data accuracy.

Tech Stack: Java, Spring Framework, Spring Boot, OOP, RabbitMQ

([Website](#))

PROJECTS

[FlameEye](#): AI-Powered Wildfire Detection & Intelligence System

- Developed a real-time wildfire detection and situational awareness platform using YOLOv8 to detect fire and smoke from live camera streams.
- Built an interactive GIS dashboard integrating NASA satellite data, multi-camera incident clustering, and physics-based wildfire spread prediction to support real-time emergency response.

Tech Stack: Python, YOLOv8, OpenCV, Computer Vision, GIS

[Academic Vocabulary Search Engine](#)

(<http://acavoca.com> | [News Article](#))

- Designed, developed, and deployed acavoca.com, a web-based academic synonym search engine that helps users discover formal alternatives for academic and professional writing.
- Independently delivered the complete project, including UI/UX design, frontend development, deployment, and content organization under the supervision of Dr. Aykut Yıldız (TED University).

Tech Stack: HTML, CSS, JavaScript

[Monitoring Online Narratives on Extremism](#)

- Developed a Python-based system to collect and analyze online news content related to extremism, radicalization, and hate speech narratives.
- Built data collection and text analysis pipelines under the supervision of a faculty member from TED University, **resulting in an [academic reference letter](#)** recognizing the project.

Tech Stack: Python, Web Scraping, NLP, Data Analysis

Elasticsearch Subtitle Search Backend

- Designed and implemented an Elasticsearch-based backend enabling fast keyword searches across video subtitles with precise timestamp retrieval.
- Developed during a backend engineering internship at Saat Teknoloji, improving search capabilities for subtitle-integrated streaming platforms.

Tech Stack: Java, Spring Boot, Elasticsearch

Student Automation System

- Developed a full-stack student management system supporting CRUD operations, user management, and basic reporting functionalities.
- Designed a modern client-server architecture using Blazor WebAssembly, .NET 9, and PostgreSQL, with containerized deployment support.

Tech Stack: C#, .NET 9, Blazor WebAssembly, PostgreSQL, HTML, CSS, JavaScript, Docker

Image Segmentation using K-Means & Gaussian Mixture Models

([Project Report](#))

- Implemented K-Means Clustering and Gaussian Mixture Models (GMM) entirely from scratch for image segmentation without relying on machine learning libraries.
- Explored unsupervised learning techniques for pixel clustering and image analysis as part of the Introduction to Machine Learning course (Final Grade: 100/100).

Tech Stack: Python, Machine Learning

Softmax Regression from Scratch

([Project Report](#))

- Implemented a custom Softmax Regression model from scratch using low-level PyTorch tensor operations, without utilizing high-level neural network modules.
- Evaluated the effects of polynomial feature expansion, learning rates, and multiple regularization techniques (Ridge, Lasso, Elastic Net) on classification performance using the Iris dataset (Final Grade: 100/100).

Tech Stack: Python, PyTorch, Machine Learning

TECHNICAL SKILLS

Programming | Java • Python • C# • C++ • C • JavaScript • PHP

Framework & Libraries | Spring Boot • .NET 9 • Blazor • Flutter • PyTorch • OpenCV

Databases | PostgreSQL • MySQL • Firebase • Cloud Firestore

AI & Data| YOLOv8 • Machine Learning • NLP • Elasticsearch

Developer Tools | Git • Docker • Linux • IntelliJ IDEA • Eclipse • VS Code

AI Tools | OpenAI • Claude • Gemini • Cursor

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Swim Team Member | TED University

Organization Team Member | TEDU Economy & Finance Society

Web Team Member | Google Developer Groups TEDU

Public Relations Team Member | Google Developer Group TEDU

Project and Competition Team Member | TEDU AI & Data Science

Vice President | Interact Club Kastamonu

Volunteer Supplier | 2017 Texas Hurricane Harvey

CERTIFICATIONS

[Huawei Cloud Developer](#) | Huawei

[HCCDA GaussDB](#) | Huawei

[Linux Fundamentals & Command Line Operations](#) | miuul

[Web Application Development with R: Shiny](#) | miuul

[Software Product Management](#) | University of Alberta (Coursera)

[Spring Boot for Beginners](#) | Amigoscode

[Modeling Software Systems using UML](#) | The Hong Kong University of Science and Technology (Coursera)

HONORS & AWARDS

[Ranked 2nd in Bilkent Swim Cup Men 100 M Backstroke](#) | Bilkent University

LANGUAGES

Turkish (Native)

English (Professional Working Proficiency / C1)

German (A2)